

Computer Vision (EE6102)

Image Foundation

2026 Fall

What is an image?

- Ideally, we think of an **image** as a 2-dimensional light intensity function, $f(x, y)$, where x and y are spatial coordinates, and f at (x, y) is related to the brightness or color of the image at that point.
- In practice, most images are defined over a rectangle.
- Continuous in amplitude (continuous-tone)
- Continuous in space: no pixels!

Digital Images and Pixels

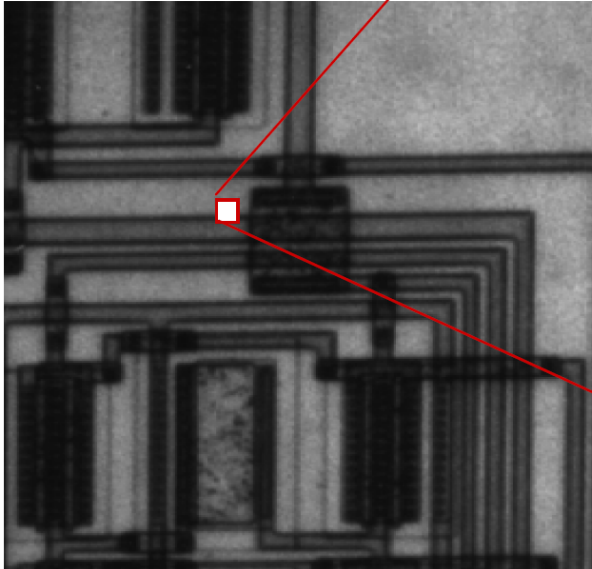
- A **digital image** is the representation of a continuous image $f(x, y)$ by a 2-d array of discrete samples.
- The amplitude of each sample is quantized to be represented by a finite number of bits. (8 bits: 0, 1, ..., 255)
- Each element of the 2-d array of samples is called a **pixel** or **pel** (from picture element)
- Pixels are point samples, without extent.
- A pixel is not:
 - Round, square, or rectangular
 - An element of an image sensor
 - An element of a display
 - But sample value at a certain position

1. A digital image is a sampled and quantized version of a continuous image.
2. Sampling discretizes spatial coordinates; quantization discretizes intensity.
3. A pixel is a point sample—not inherently a square physical object.

A digital image is represented by umbers

What do you think is in this box?

272 pixels



280 pixels

128	125	107	105	110	118	116	114	110
121	122	115	108	106	107	116	116	107
110	114	112	107	105	103	106	106	100
100	96	100	99	94	94	101	101	89
85	82	81	80	76	75	80	82	72
58	58	56	54	53	52	51	49	45
41	41	41	39	39	38	36	35	33
43	43	42	43	41	41	41	43	40
60	60	59	59	60	59	59	58	56

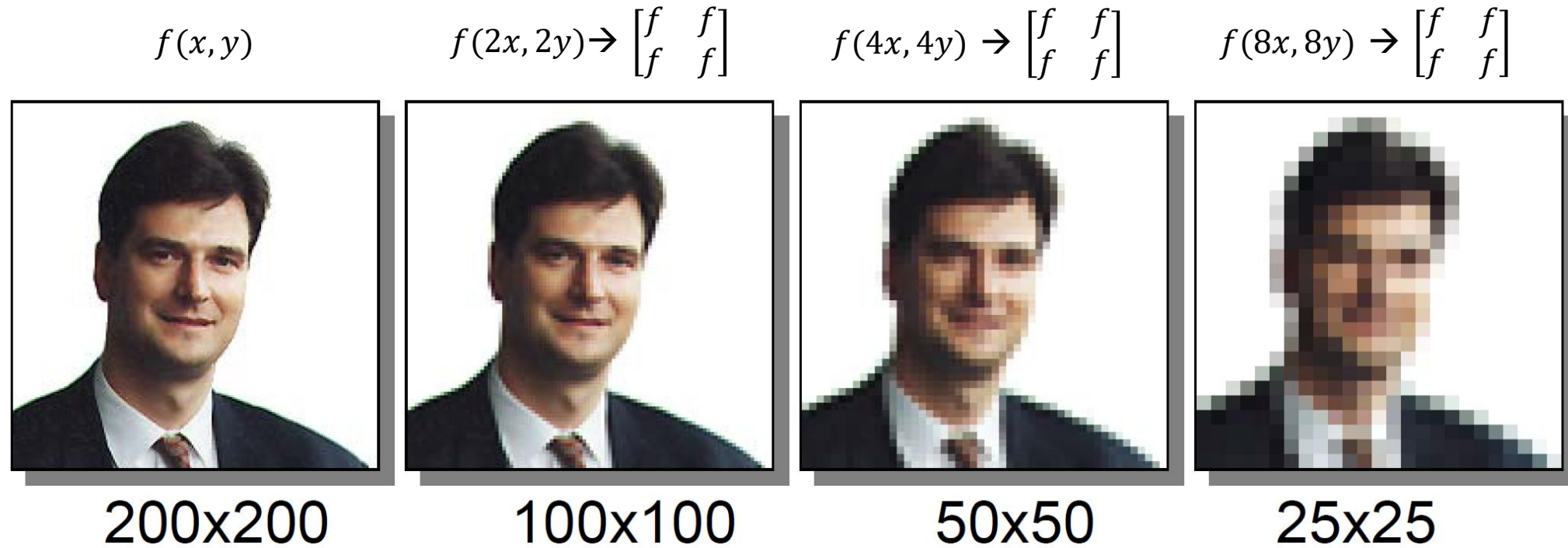
- Pixel = “picture element”
- Represents brightness at one point

A digital image can be represented by a matrix

$$\mathbf{f} = \begin{matrix} & \xrightarrow{x} & & \\ \left[\begin{array}{cccc} f(0,0) & f(1,0) & \cdots & f(N-1,0) \\ f(0,1) & f(1,1) & \cdots & f(N-1,1) \\ \vdots & \vdots & & \vdots \\ f(0,L-1) & f(1,L-1) & \cdots & f(N-1,L-1) \end{array} \right] & \downarrow y & \end{matrix}$$

- The pixel values $f(x, y)$ are sorted into the matrix in *natural* order, with x corresponding to the column and y to the row index. Python or Matlab uses this convention.
- This results in $f(x, y) = f_{yx}$, where f_{yx} denotes an individual element in common matrix notation.
- For a color image, $\mathbf{f}$ might be one of the components.

Image Size and Resolution



- These images were produced by simply picking every n -th sample horizontally and vertically and replicating that value $n \times n$ times
- We can do better
 - Prefiltering before subsampling to avoid **aliasing**
 - Smooth interpolation

Image Size and Resolution

$$f(x, y)$$



$$f(2x, 2y)$$



$$f(4x, 4y)$$

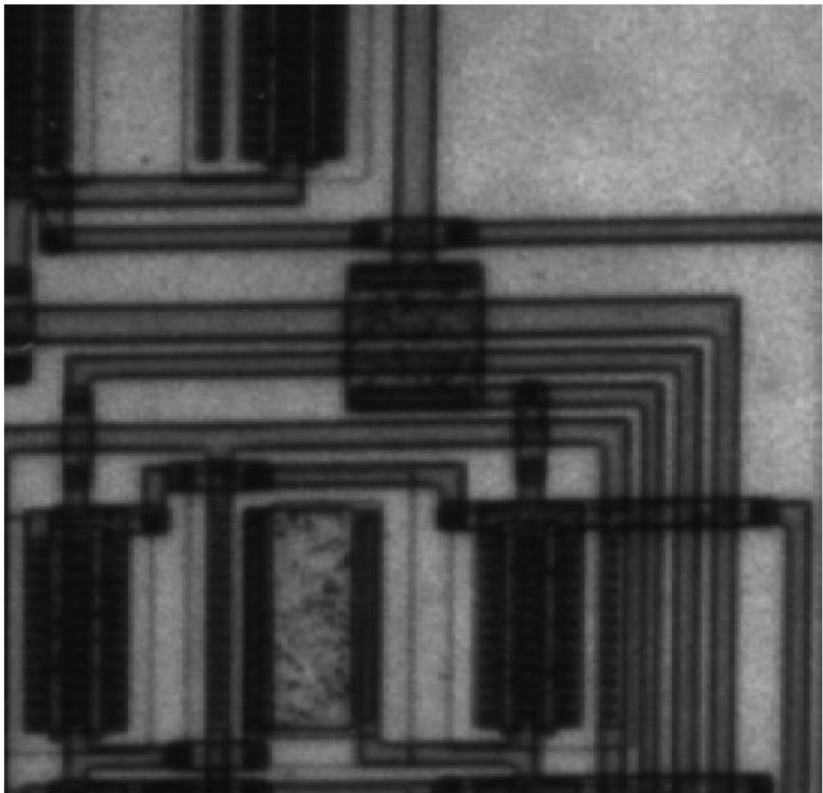


$$f(8x, 8y)$$



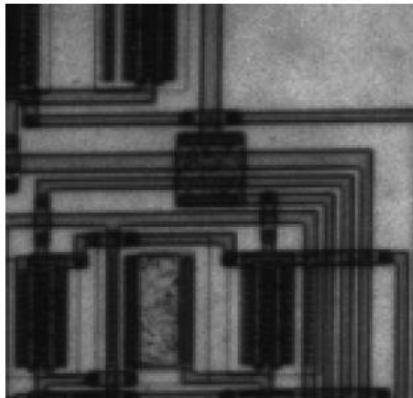
Images of Different Sizes

272 pixels



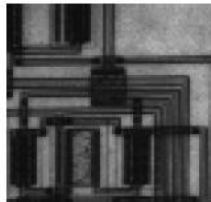
280 pixels

136



140

66



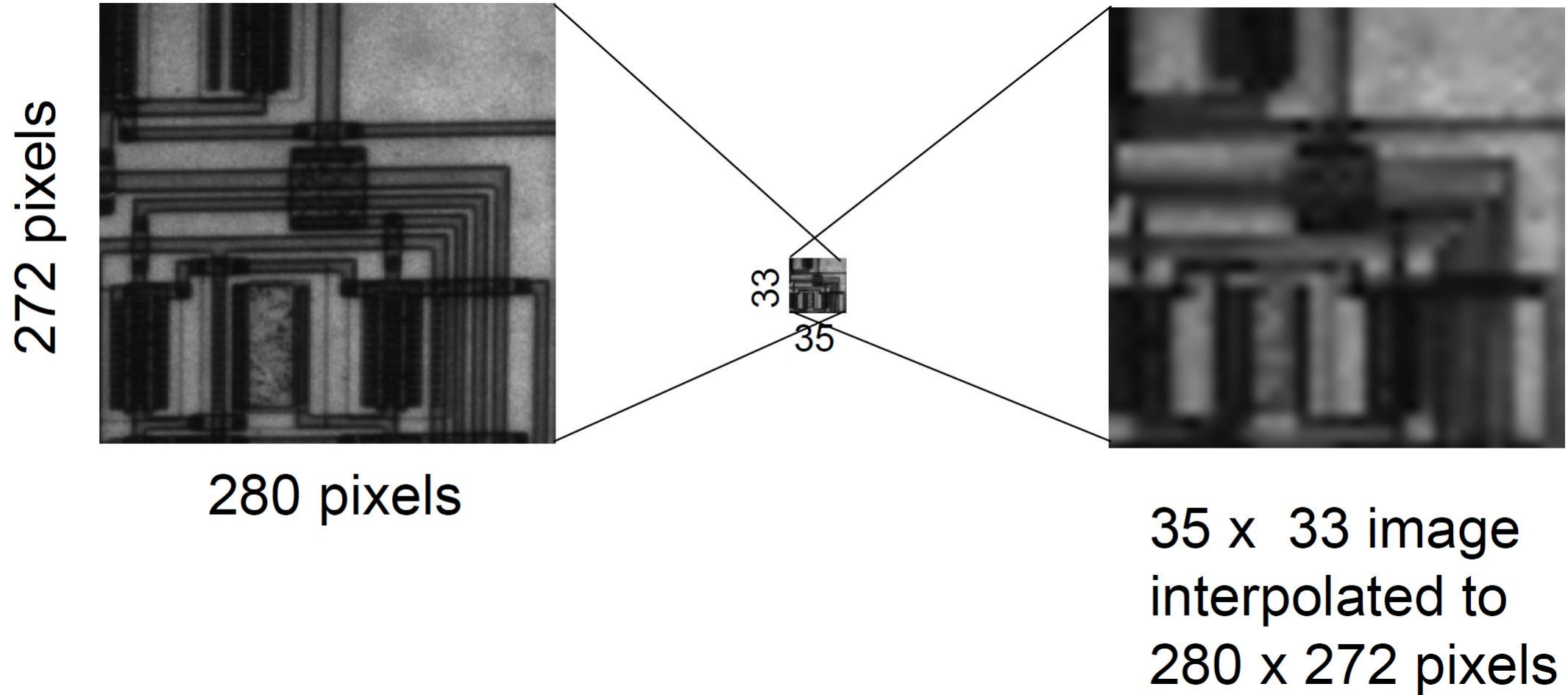
70

33

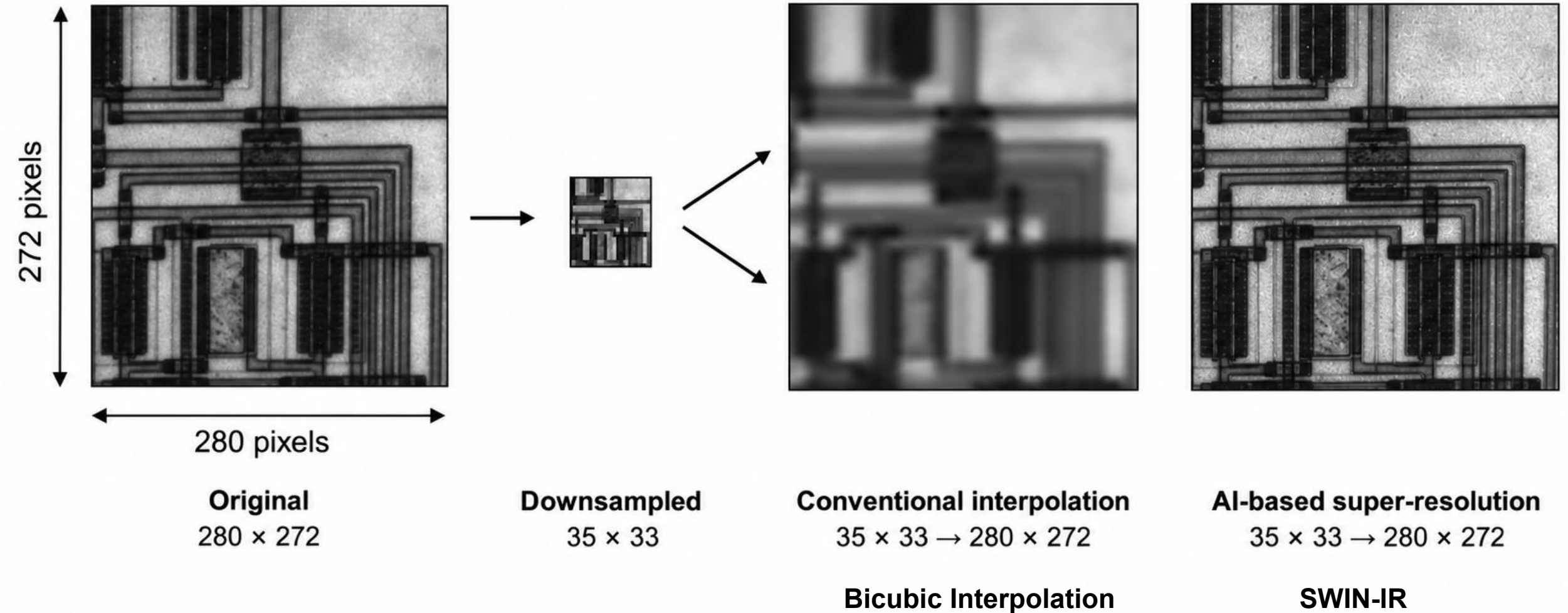


35

Fewer Pixels Mean Lower Spatial Resolution



How AI Improves Enlarged Low-Resolution Images



Bicubic interpolation estimates missing pixels from neighboring values, while AI-based super-resolution predicts likely high-frequency details from learned image patterns.

Color Components

$$f(x, y) = \{R(x, y), G(x, y), B(x, y)\}_{H \times W \times 3}$$

Monochrome image



$$R(x, y) = G(x, y) = B(x, y)$$



Red $R(x, y)$



Green $G(x, y)$

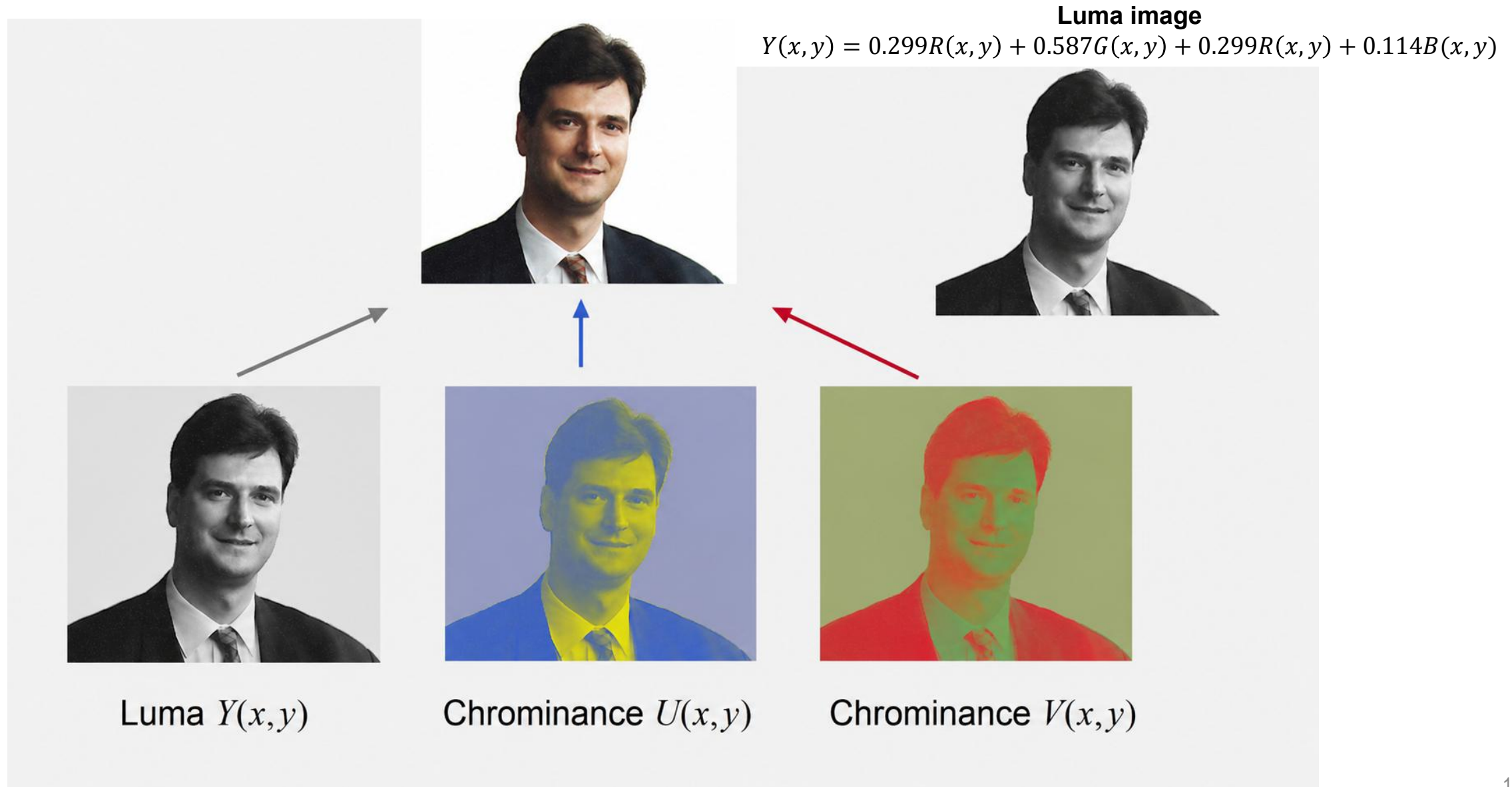


Blue $B(x, y)$

The R, G, and B light components are additively mixed by the display and perceived by the human visual system as a single color.

- (255,0,0) → red
- (0,255,0) → green
- (0,0,255) → blue
- (255,255,0) → red + green light → yellow
- (255,255,255) → red + green + blue light → white

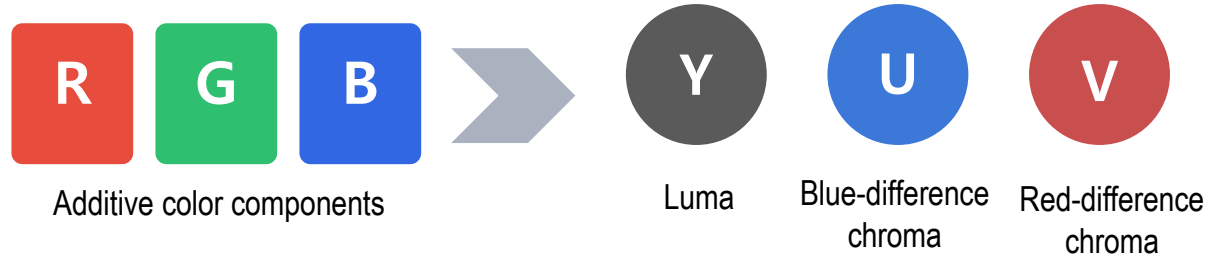
YCbCr Components



RGB and YCbCr: Perception and Chroma Subsampling

YUV separates brightness from color so that chroma can be represented more efficiently.

RGB → YUV



Conceptual transform

$$Y = 0.299R + 0.587G + 0.114B$$

$$C_b \propto B - Y$$

$$C_r \propto R - Y$$

Y carries most structural detail; U and V mainly carry color differences.

Why YCbCr?

Separate luminance from chrominance

Brightness and color are represented independently.

Matches human visual sensitivity

The eye is more sensitive to fine luminance detail than to fine color detail.

Enables chroma subsampling

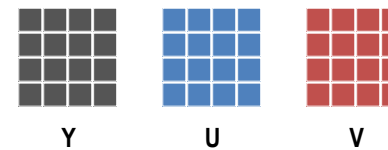
Cb/Cr can be stored at lower spatial resolution with relatively small perceptual loss.

Improves compression efficiency

This principle is widely used in image/video coding and transmission.

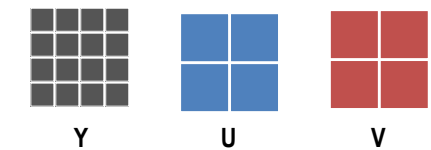
4:4:4 vs 4:2:0

4:4:4



Y, U and V all have the same spatial resolution.

4:2:0

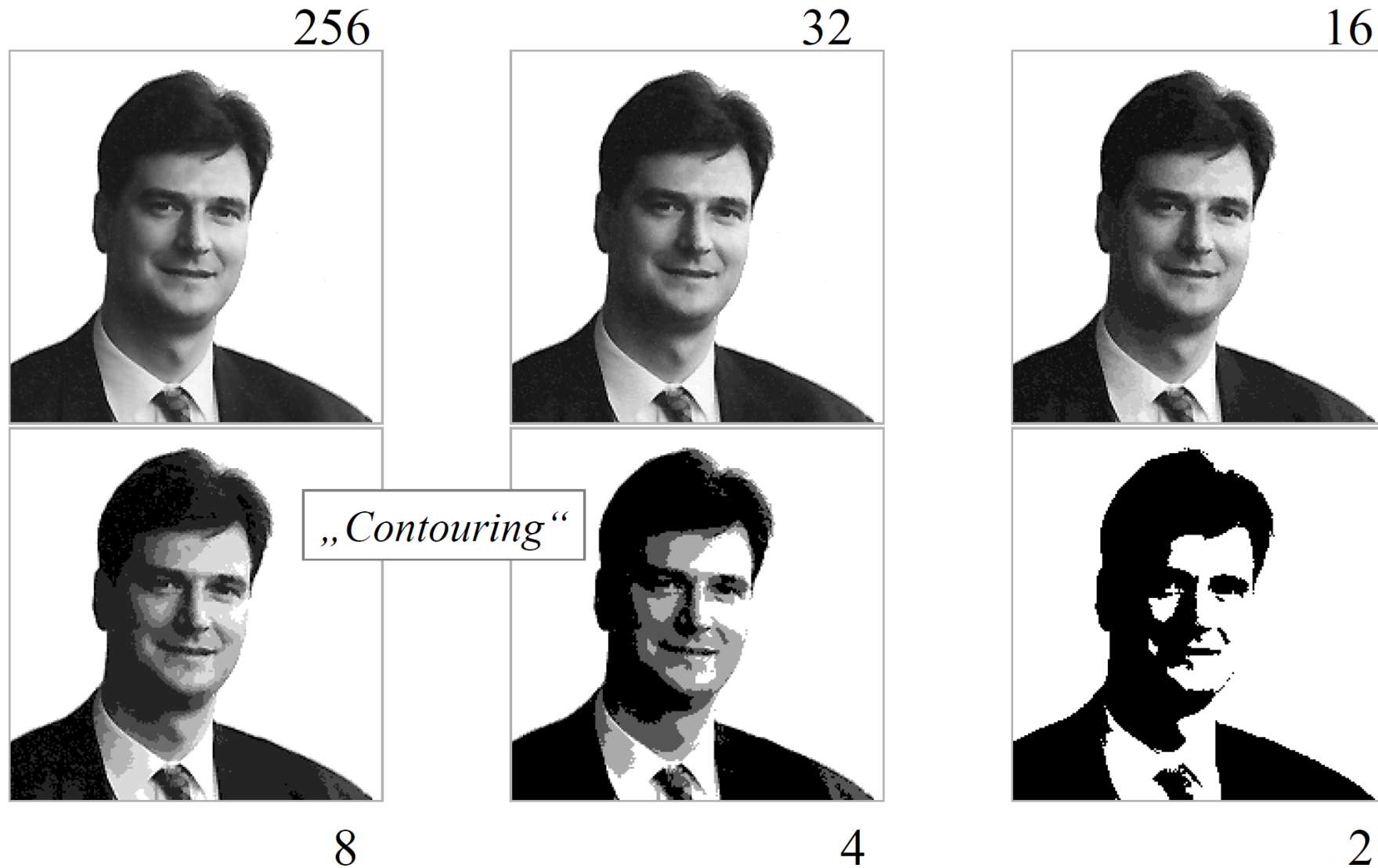


Y stays full-resolution; U and V use 1/2 width × 1/2 height (1/4 as many samples each).

If Y, U and V are shown at the same resolution, the representation is 4:4:4 — not 4:2:0.

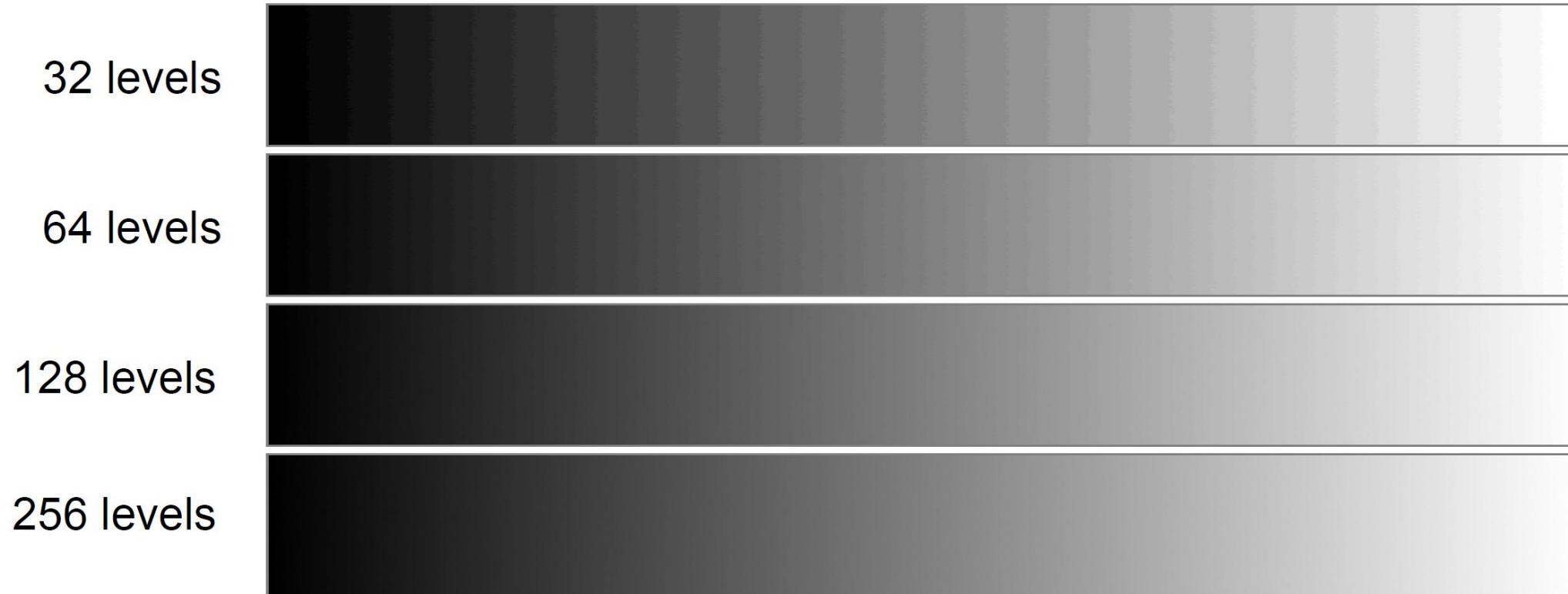
- 4:2:0 significantly reduces data size while preserving perceived visual quality, because the human visual system is less sensitive to fine chroma detail.
- If 4:4:4 has the same number of samples as RGB, why do we still convert RGB to YCbCr?

Different Numbers of Gray Levels



How many gray levels are required

- Contouring is most visible for a ramp



Digital images typically are quantized to 256 gray levels

Gray Levels, Bit Depth, and Dynamic Range

Two different questions: how finely do we sample intensity, and how wide an intensity range can we represent?

1 Gray levels = intensity resolution $L = 2^b$

Fewer gray levels → visible banding / false contouring

32 levels 5-bit



64 levels 6-bit



256 levels 8-bit



More levels make the ramp look smoother, but they do not by themselves define the scene's dynamic range.

2 Dynamic range = span from dark to bright

$$DR = S_{\max} / S_{\min} \quad \bullet \quad DR \text{ (stops)} = \log_2(S_{\max} / S_{\min})$$

where $S_{\min}$ is the darkest detectable non-zero signal

Wide dynamic range



shadow detail

highlight detail

Insufficient range



clipped to black

clipped to white

Key distinction

- Bit depth tells us how many intensity codes are available.
- Dynamic range tells us how far apart the darkest and brightest reproducible signals can be.

Takeaway: More gray levels reduce banding; wider dynamic range preserves detail in both shadows and highlights.

Dynamic Range and Visual Quality

- **Dynamic Range (DR):** Range between the darkest and brightest intensities that an imaging system can capture or display.

$$DR = I_{max}/I_{min}$$

$$DR_{dB} = 20 \log_{10}(I_{max}/I_{min})$$

Narrow Dynamic Range	Wide Dynamic Range
Shadows → crushed to black	More shadow detail
Highlights → clipped to white	More highlight detail
Limited contrast range	Richer range of brightness
Loss of information	Better preservation of scene information

Wider Dynamic Range leads to more natural and perceptually richer images

- Better visibility in both dark and bright regions
- Greater perceived contrast and depth
- More realistic representation of high-contrast scenes
- Particularly noticeable for sunlight, reflections, night scenes, and bright highlights

Dynamic Range and Visual Quality

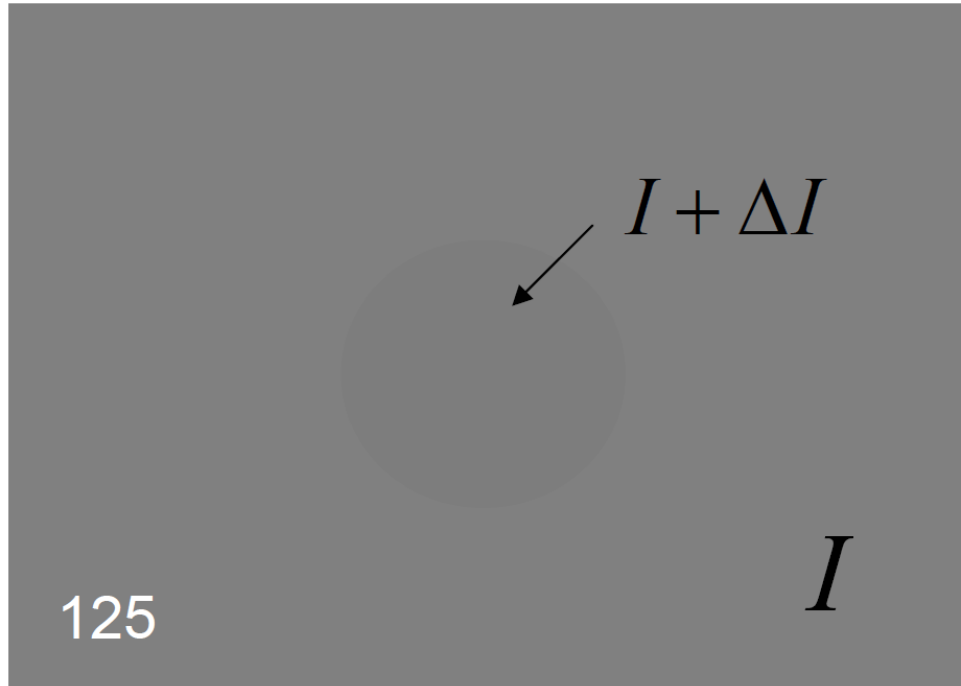
Service / Standard	SDR	HDR
Traditional Broadcasting (ATSC 1.0)	8-bit	Not supported
ATSC 3.0 / Next-Generation TV	8-bit or 10-bit	10-bit
YouTube	Typically 8-bit	10-bit or 12-bit
Netflix	Typically 8–10-bit	Mainly 10-bit or higher; 12-bit may be used for Dolby Vision production

Storage Requirements for Digital Images

- Image $L \times N$ pixels, 2^B gray levels, c color components
- $\text{Size} = L \times N \times B \times c$
 - *Example: $L=N=512$, $B=8$, $c=1$ (i.e., monochrome)*
 - *Size = 2,097,152 bits (or 256 kByte)*
 - *Example: $L \times N=1024 \times 1280$, $B=8$, $c=3$ (24 bit RGB image)*
 - *Size = 31,457,280 bits (or 3.75 MByte)*

Brightness Discriminant Experiment

- Can you see the circle?



The background has luminance I , and the circle is slightly brighter by ΔI

Note: I is the luminance measured in cd/m^2 (nit)

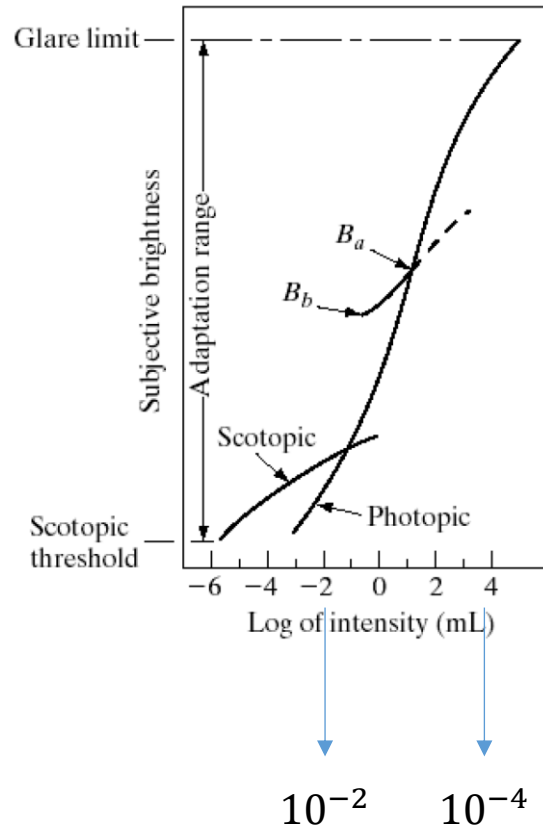
- Visibility threshold

$$\frac{\Delta I}{I} \approx \text{const.} \approx 1\sim 2\%$$



Weber

Bright Adaptation of the Human Eye



Perceived brightness (intensity perceived by visual system) is a logarithmic function of the incident light intensity

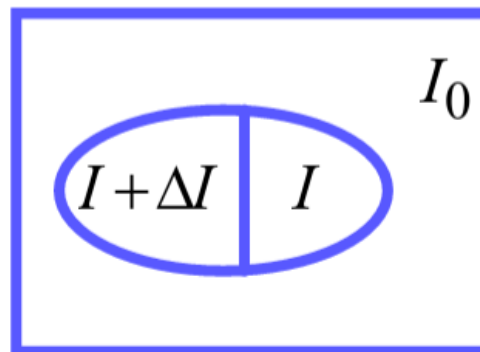
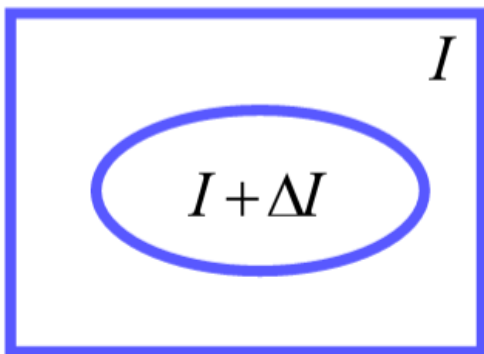
Enormous dynamic range! In photopic vision alone $\sim 10^6$

Within each adaptation level the range of intensities that we can simultaneously discriminate is rather small

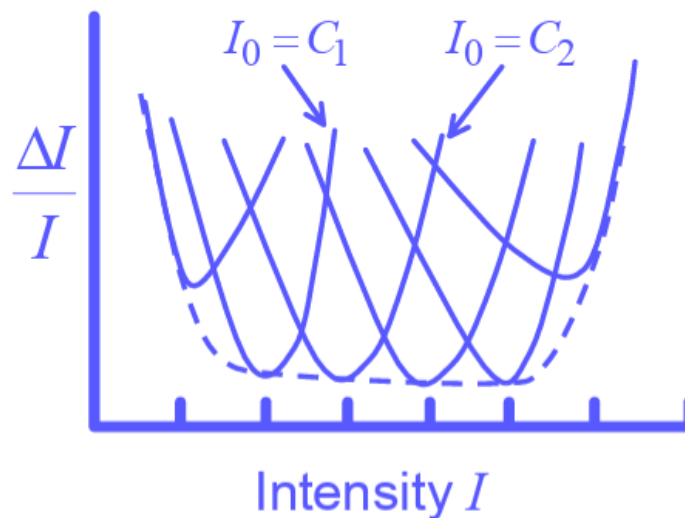
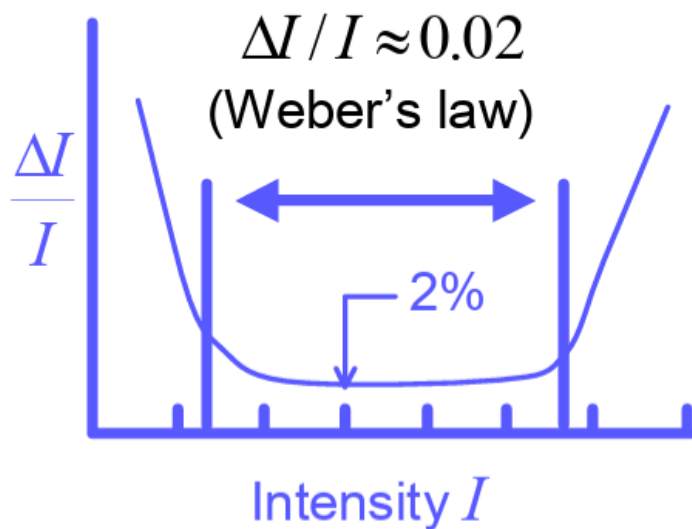
The human eye covers an enormous dynamic range by continuously adapting its sensitivity to the ambient luminance level.

Weber's Law

Just noticeable difference :
(gray light)



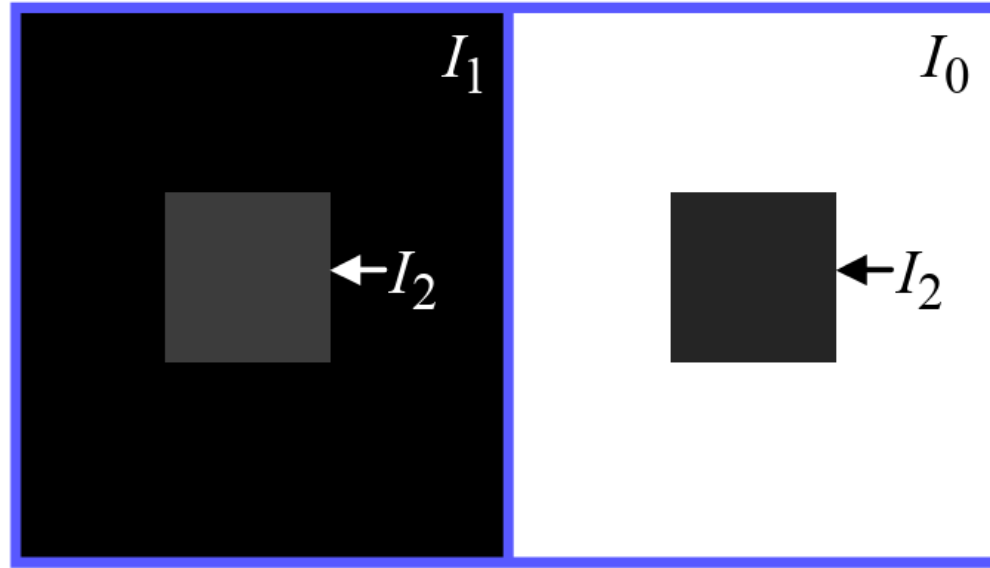
$$\Delta I / I \approx 0.02 f(I - I_0)$$



The minimum discrimination threshold shifts with the adaptation luminance I_0

Experiments of Weber more than 100 years ago

Simultaneous Contrast



Same intensity, different perceived brightness due to different surroundings.

- Simultaneous contrast: the perceived intensity of an object depends on the intensity of the background “contrasting colors enforce each other”
- This is another obvious example that the eye adapts itself to the global brightness of the surrounding

Contrast with 8 Bits According to Weber's Law

- Assume that the luminance difference between two successive representative levels is just at visibility threshold

$$\frac{I_{max}}{I} = (1 + const.)^{255}$$

- For $const. = 0.01 \sim 0.02$, $\frac{I_{max}}{I} = 13 \sim 156$
- Typical display contrast
 - Cathode ray tube (CRT): 100:1
 - Print on paper: 10:1
- Suggests uniform quantization in the $\log(I)$ domain

Contrast with 8 Bits According to Weber's Law (FYI)

From **Weber's law**,

$$\frac{\Delta I}{I} = k$$

the next distinguishable brightness level must increase by a fraction k of the current intensity.

If the current level is I_n ,

$$\Delta I = kI_n$$

so the next level is

$$\begin{aligned} I_{n+1} &= I_n + \Delta I \\ &= I_n + kI_n \\ &= (1 + k)I_n \end{aligned}$$

Therefore, the successive intensity levels become

$$\begin{aligned} I_1 &= (1 + k)I_0 \\ I_2 &= (1 + k)I_1 = (1 + k)^2 I_0 \\ I_3 &= (1 + k)^3 I_0 \end{aligned}$$

and, in general,

$$I_n = (1 + k)^n I_0$$

For an **8-bit image**, there are **256 intensity levels**, indexed as

$$0, 1, \dots, 255$$

If the minimum intensity is

$$I_{\min} = I_0$$

and the maximum intensity is

$$I_{\max} = I_{255},$$

then

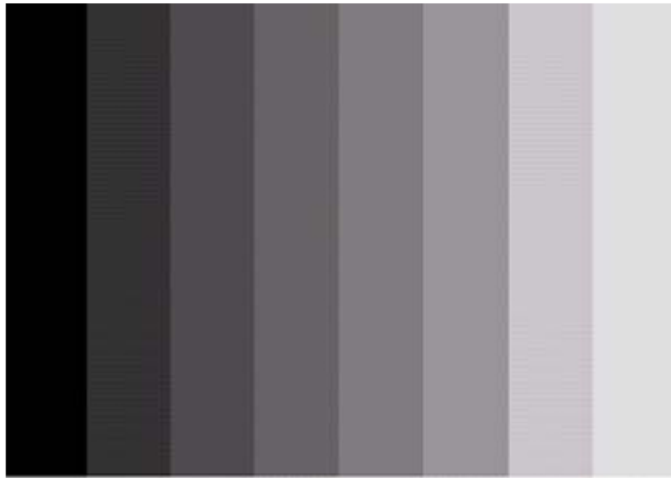
$$I_{\max} = (1 + k)^{255} I_{\min}$$

Therefore,

$$\boxed{\frac{I_{\max}}{I_{\min}} = (1 + k)^{255}}$$

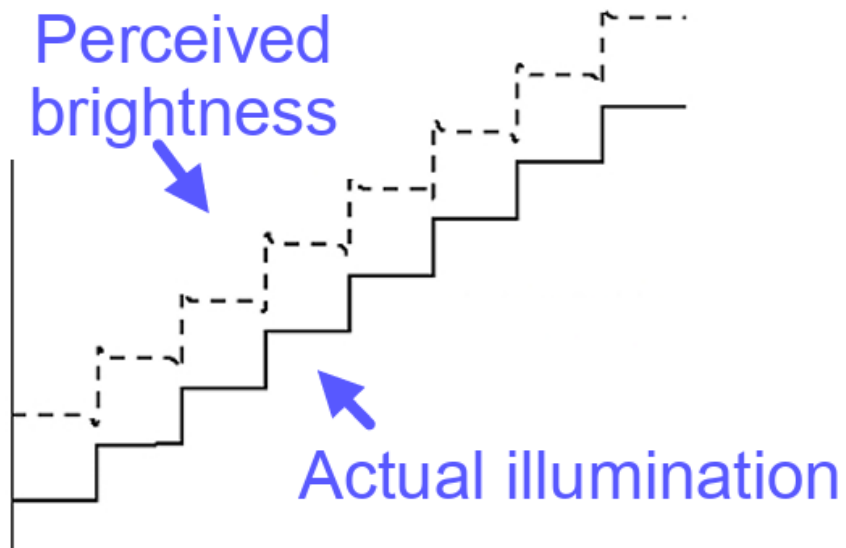
The key point is that **256 intensity levels contain 255 intervals (steps)**, which is why the exponent is **255, not 256**.

Mach Bands (The human visual system enhances contrast near intensity boundaries)



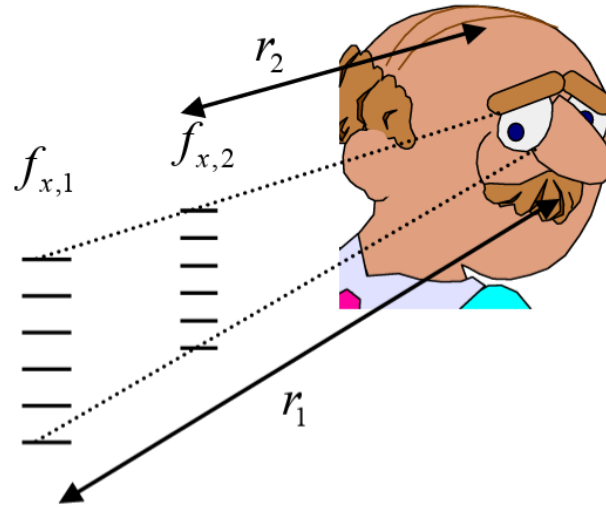
← Stripes of constant intensity

The human visual system tends to undershoot or overshoot around the boundary of the regions



This phenomenon was first discovered by Ernst Mach in 1865

Perceived Spatial Frequency



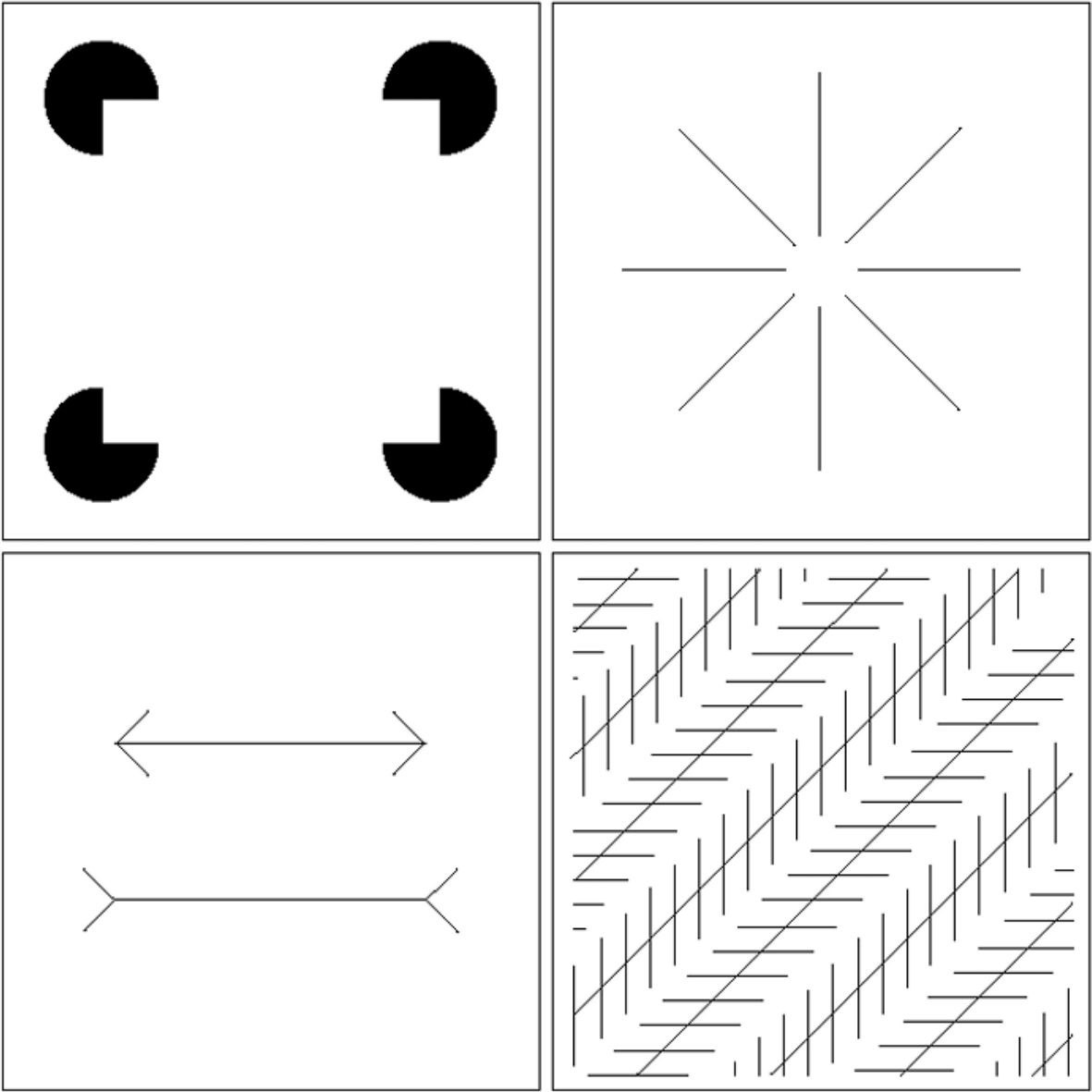
$f_{x,i}$: spatial frequency in cycle.per meter

r_i : viewing distance in meters

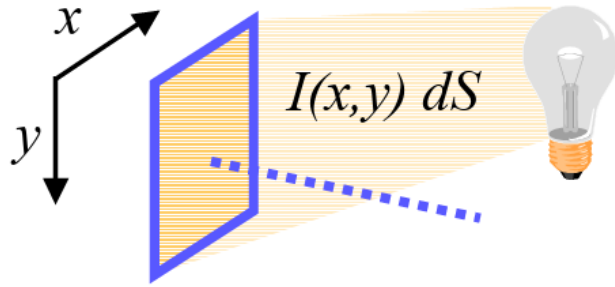
Spatial frequency perceived by the human eye depends on both the **physical frequency of the pattern** and the **viewing distance**.

- Two patterns look as if have the same frequency if $\frac{f_{x,1}}{f_{x,2}} = \frac{r_2}{r_1}$
- $f_x r$ is a good measure for the perceived frequency (cycle per degree)

Optical Illusion



Light Intensity



Remark: visible light

$\lambda \approx 550$ nm (order of magnitude)

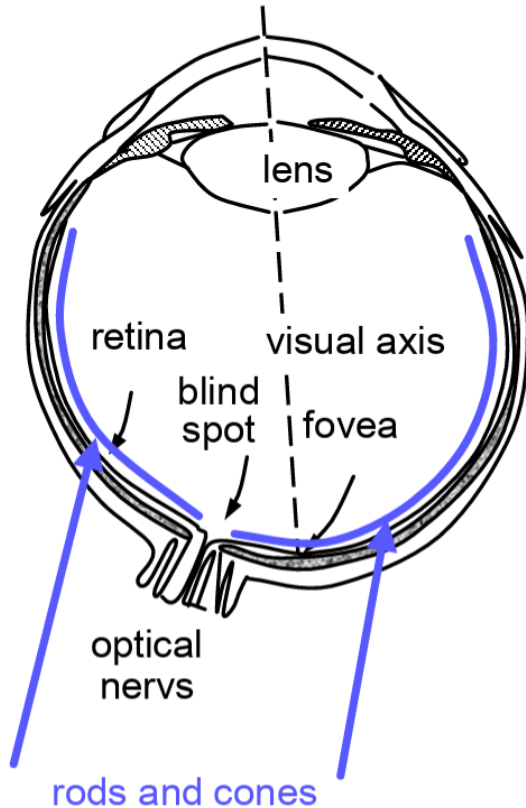
$f \approx 500$ THz (order of magnitude)

- Light intensity: $I(x, y)dS$ is the light power that is incident on an element part of the surface dS (unit: watt/m²)
- Light can be decomposed into a color spectrum

$$I(x, y) = \int_{-\infty}^{\infty} \underbrace{I(x, y; f) df}_{\substack{\text{Frequency spectrum} \\ \text{(independent of the transmission} \\ \text{medium)}}} = \int_{-\infty}^{\infty} \underbrace{I(x, y; c/\lambda) |df/d\lambda| d\lambda}_{\substack{\text{Wavelength spectrum} \\ \text{(used in light technology)}}} = \int_{-\infty}^{\infty} \underbrace{I'(x, y; \lambda) d\lambda}_{\substack{\text{Wavelength spectrum} \\ \text{(used in light technology)}}}$$

In empty space: $f\lambda = c$ with c the speed light $|df/d\lambda| = c/\lambda^2$

Structure of the Human Eye



- **Light receptors**

- Two types: cones and rods
- Highest density of cones is in fovea
- 100,000,000 receptors in the eye but only 800,000 nerves

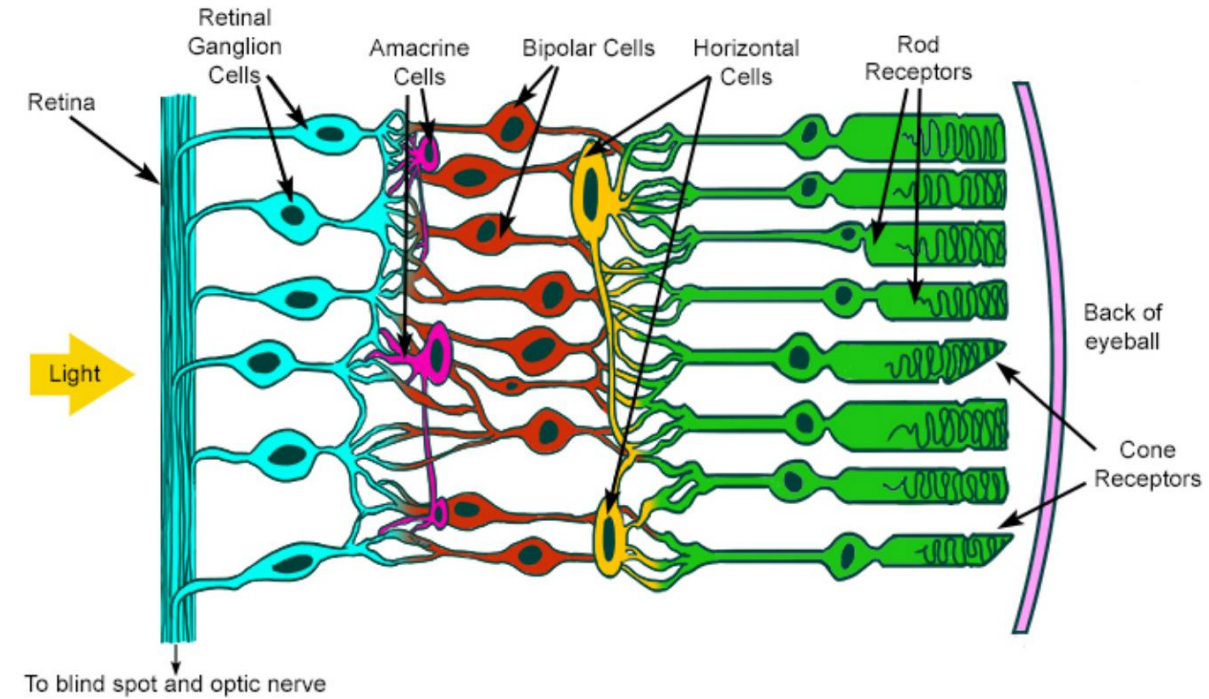
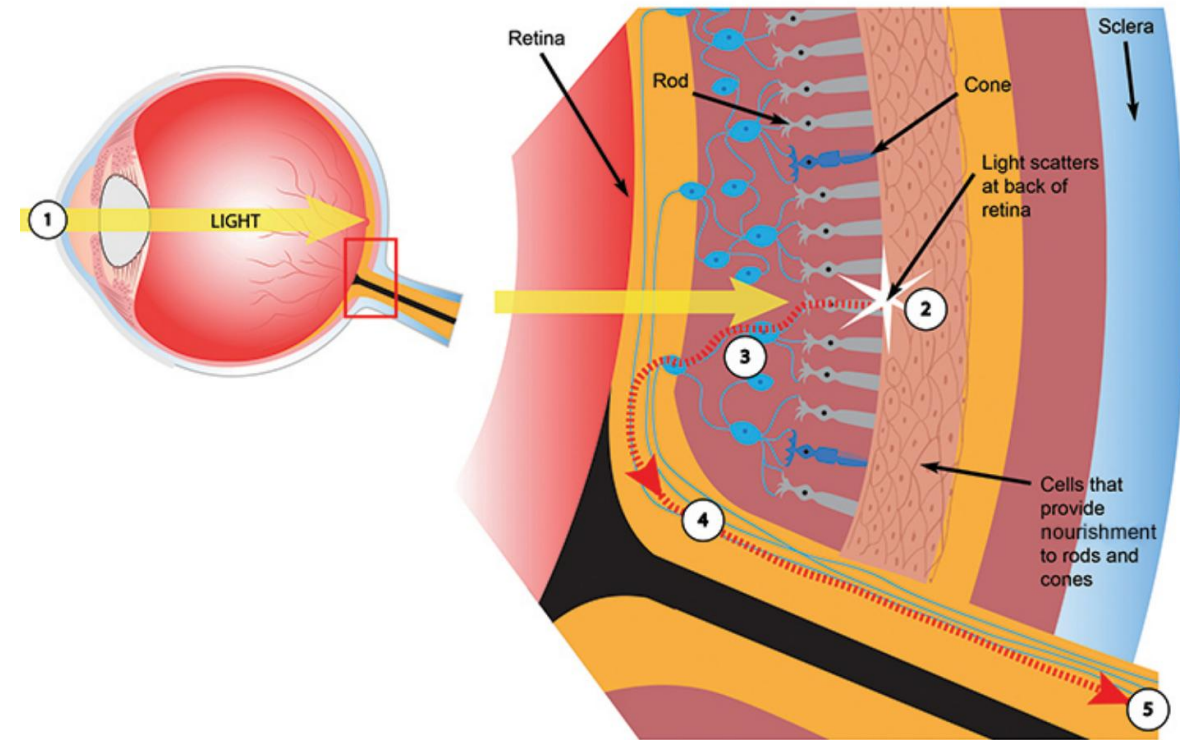
- **Cones**

- Amount: 6,000,000
- Photopic(bright-light) vision
- 3 types, sensitive for: (1) blue, (2) green, (3)yellow green to red light

- **Rods**

- Amount: 100,000,000
- Much more sensitive
- 1 type → no color distinction

Retina, Cones and Rods



Absorption Spectra of the Cone Pigments

- These absorption spectra show the percent of the light of a given frequency that is absorbed by different types of cones
- This light is not fully transformed in a nerve signal
 - The absorption spectra give only a rough idea about the **spectral sensitivity** of three different types of cones

